CS & IT ENGINEERING

DIGITAL LOGIC



Combinational circuit

Lecture No. 10

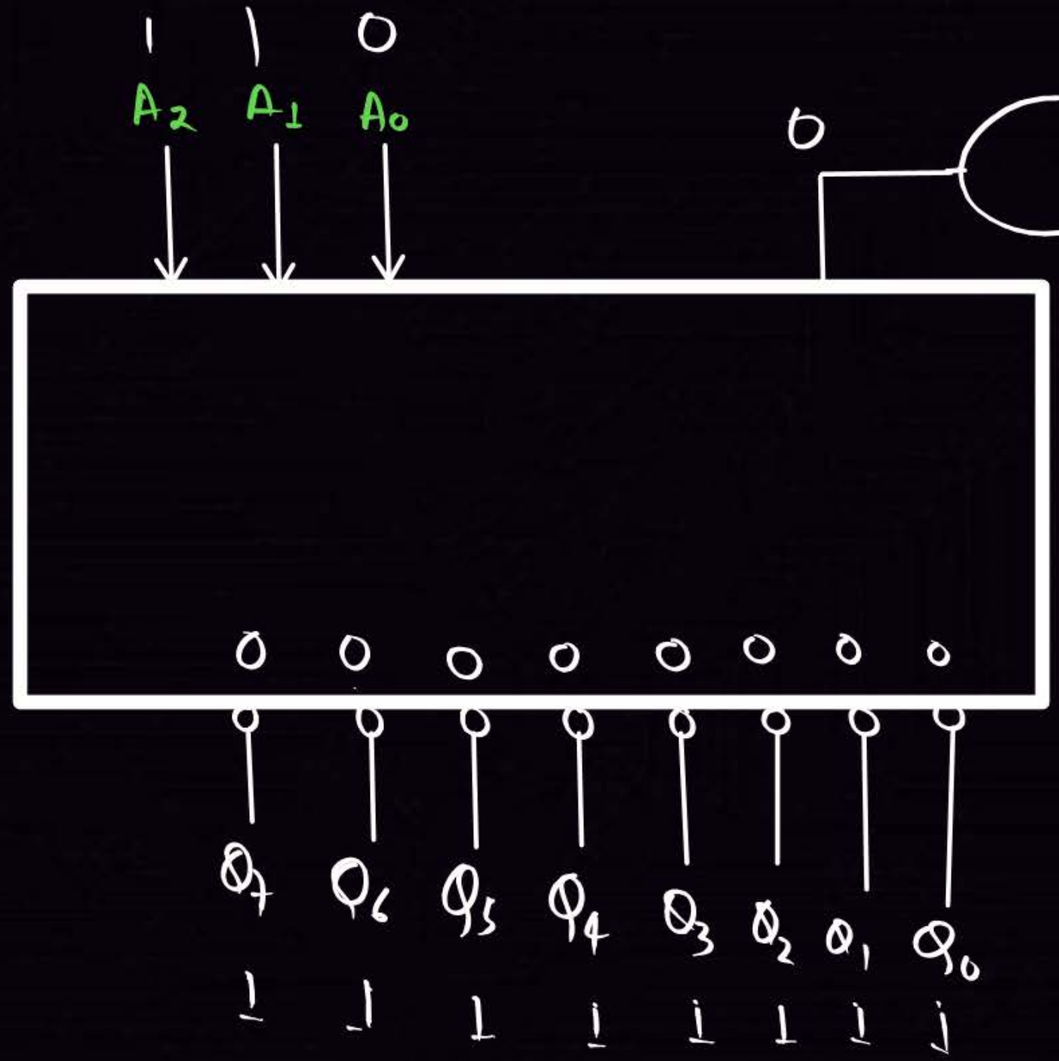


By- CHANDAN SIR



01 Question Practice

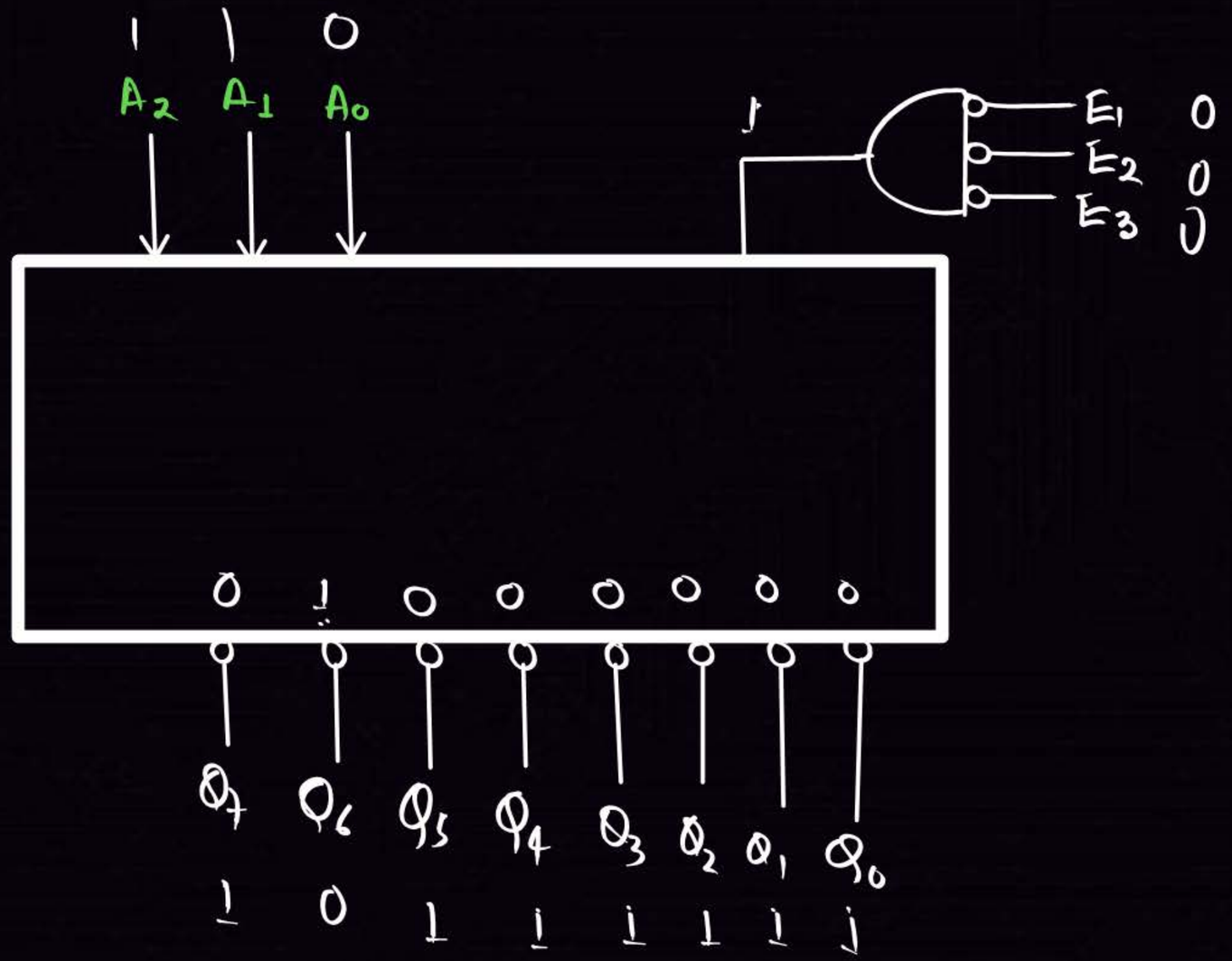
02 Discussion



(A) ✓ 1 1 1 1 1 1 1 1

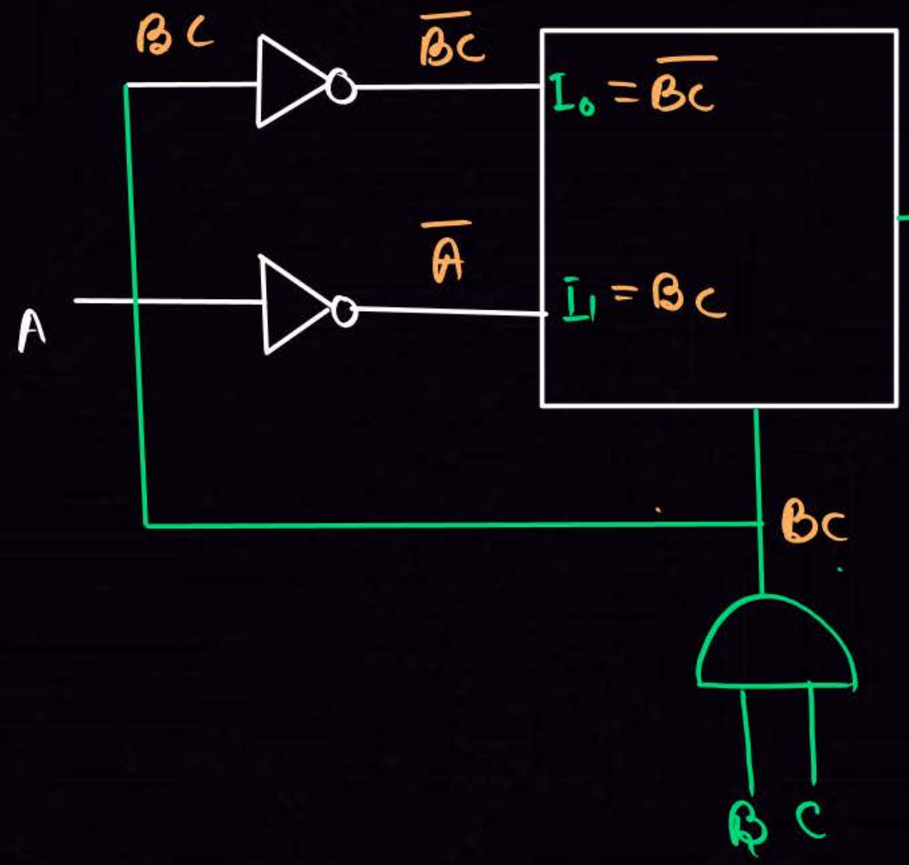
(B) 1 0 1 1 1 1 1 1

It is not enable



(A) 1 1 1 1 1 1 1 1

(B) 1 0 1 1 1 1 1 1



$\overline{BC} \cdot \overline{BC} + \overline{A} \cdot BC$

$\overline{BC} + \overline{A}BC$

$BC = X$

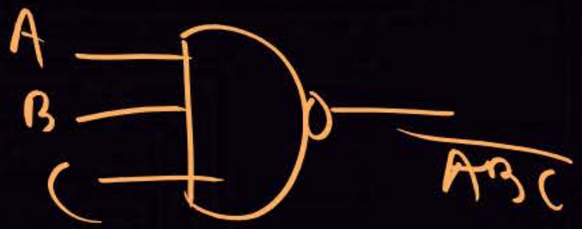
$\overline{X} + \overline{A}X$

$(\overline{X}+X)(\overline{A}+\overline{X})$

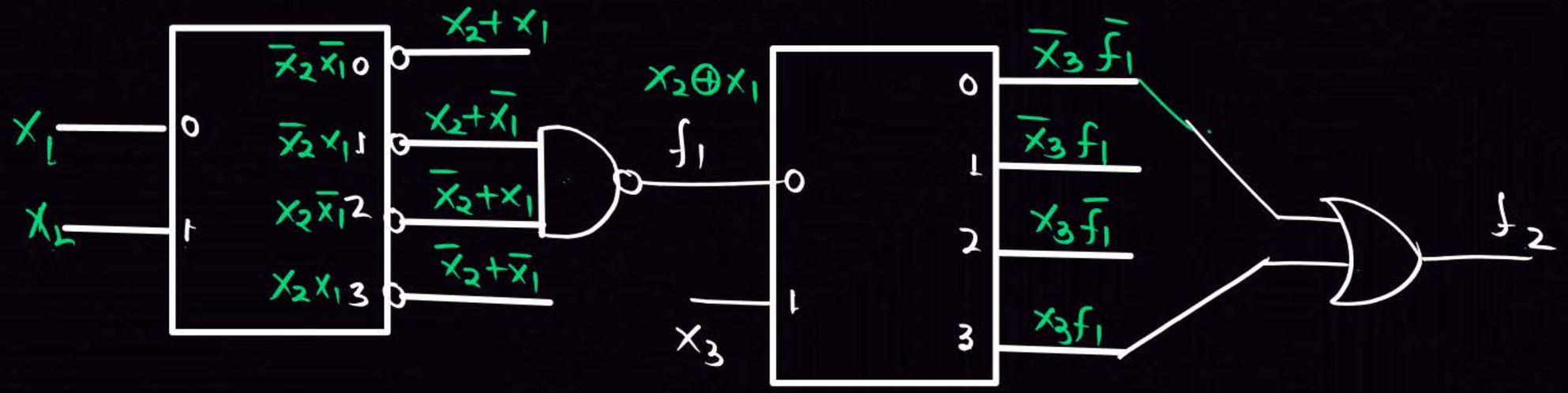
$\overline{A}+\overline{X}$

$\overline{A \cdot X}$

$\overline{A \cdot BC}$



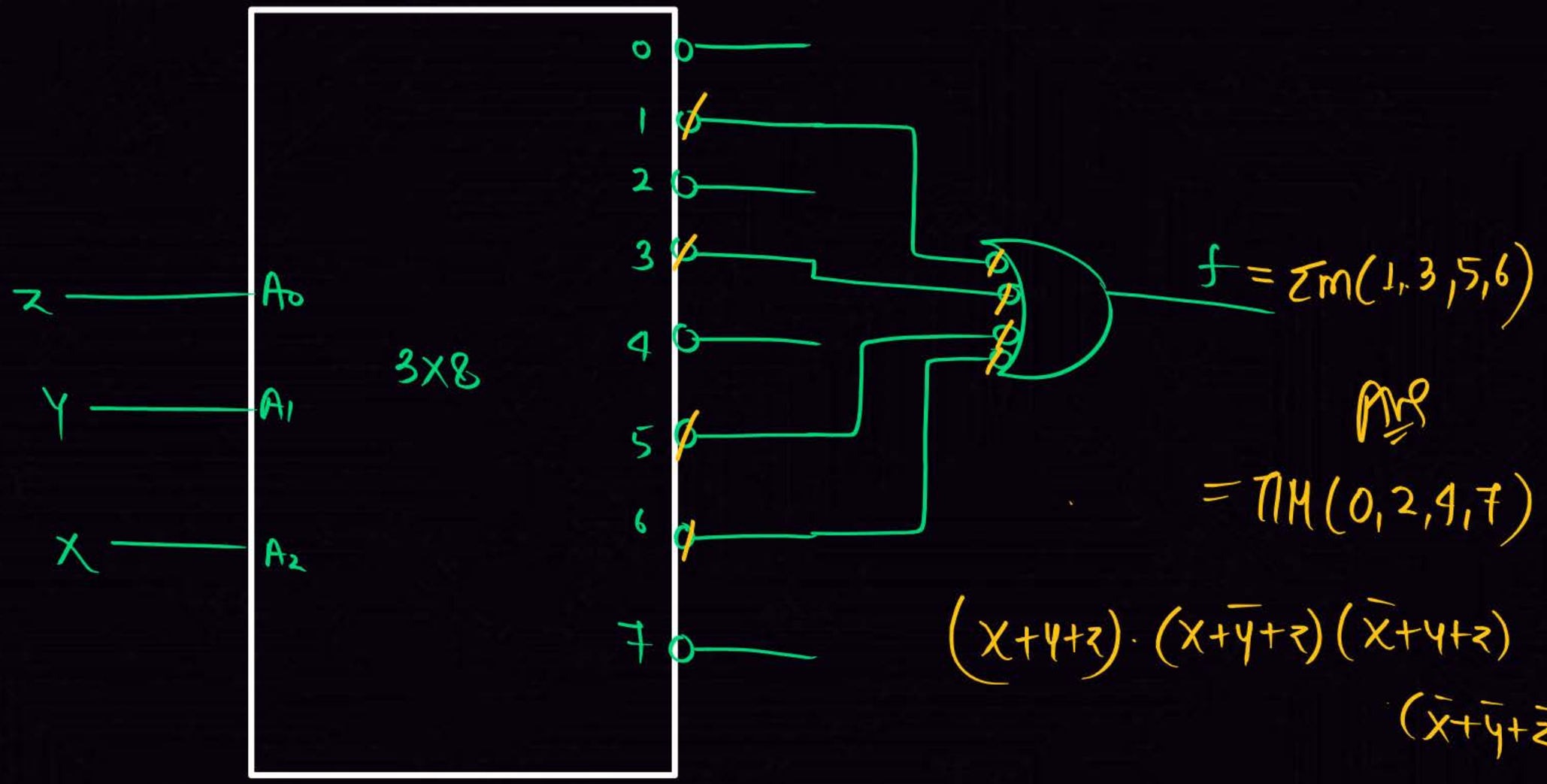
$$f_1 = \overline{(x_2 + \bar{x}_1)\cdot(\bar{x}_2 + x_1)} = \overline{x_2 \odot x_1} = x_2 \oplus x_1$$



$$f_2 = \bar{x}_3\bar{f}_1 + x_3 f_1 = x_3 \odot f_1$$

$$= x_3 \odot (x_2 \oplus x_1)$$

Q.



Ans

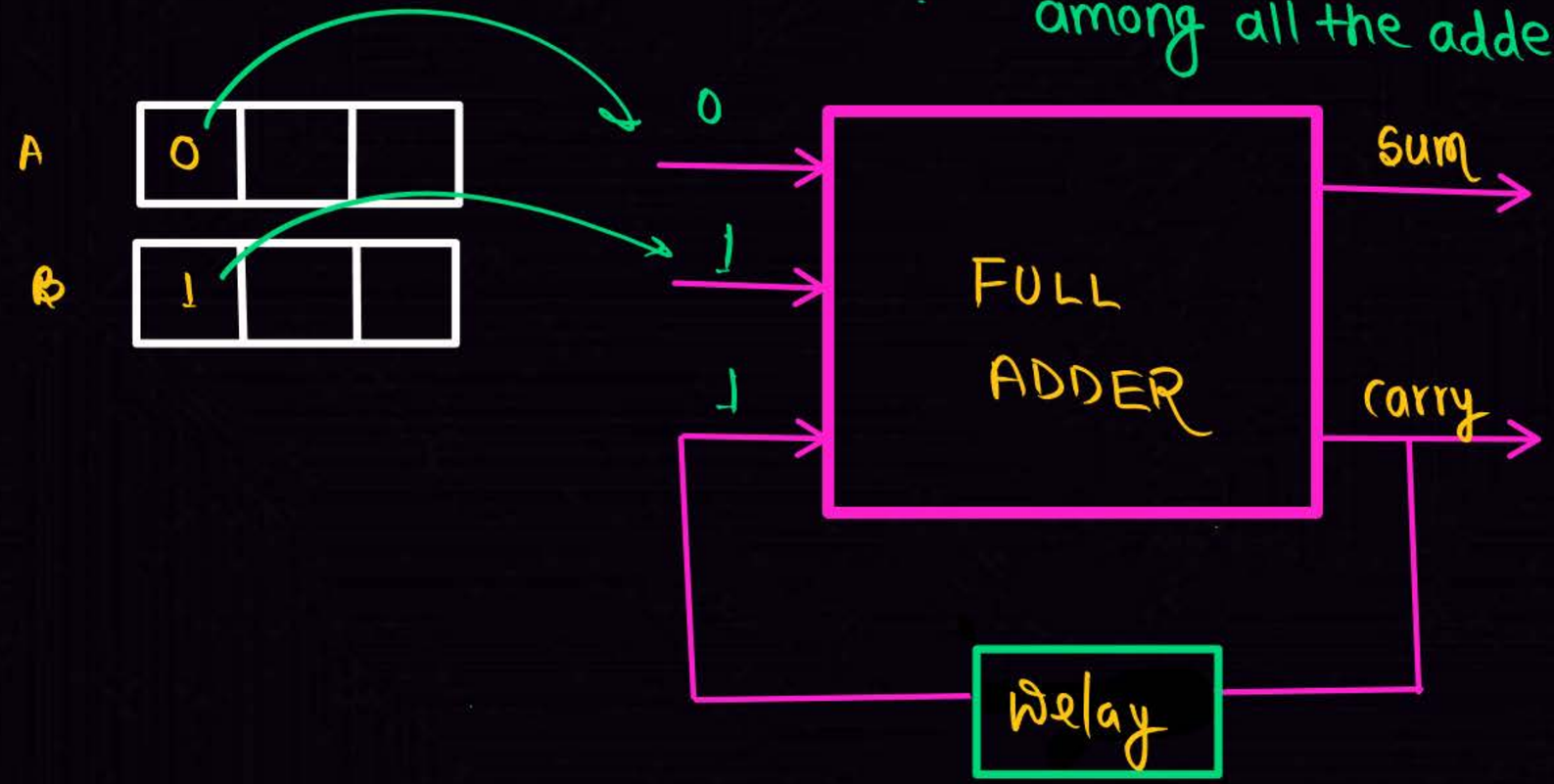
$= \Pi M(0,2,4,7)$

$(X+Y+Z)\cdot(X+\bar{Y}+Z)(\bar{X}+Y+Z)$

$(\bar{X}+\bar{Y}+\bar{Z})$

SERIAL ADDER

→ It is the slowest adder among all the adder.



Carry: 1 0 0 1

Total time for operation

$$T = n \cdot T_{FA}$$

$n \rightarrow$ no. of bits

$T_{FA} \rightarrow$ Delay of FA.

PARALLEL ADDER

4 bit parallel adder

C_3 C_2 C_1

A_3 A_2 A_1 A_0

B_3 B_2 B_1 B_0

C_4 S_3 S_2 S_1 S_0

$(N-1)\{2HA+1OR\}+1HA$

$(2N-2)HA+(N-1)OR+1HA$

$(2N-1)HA+(N-1)OR$

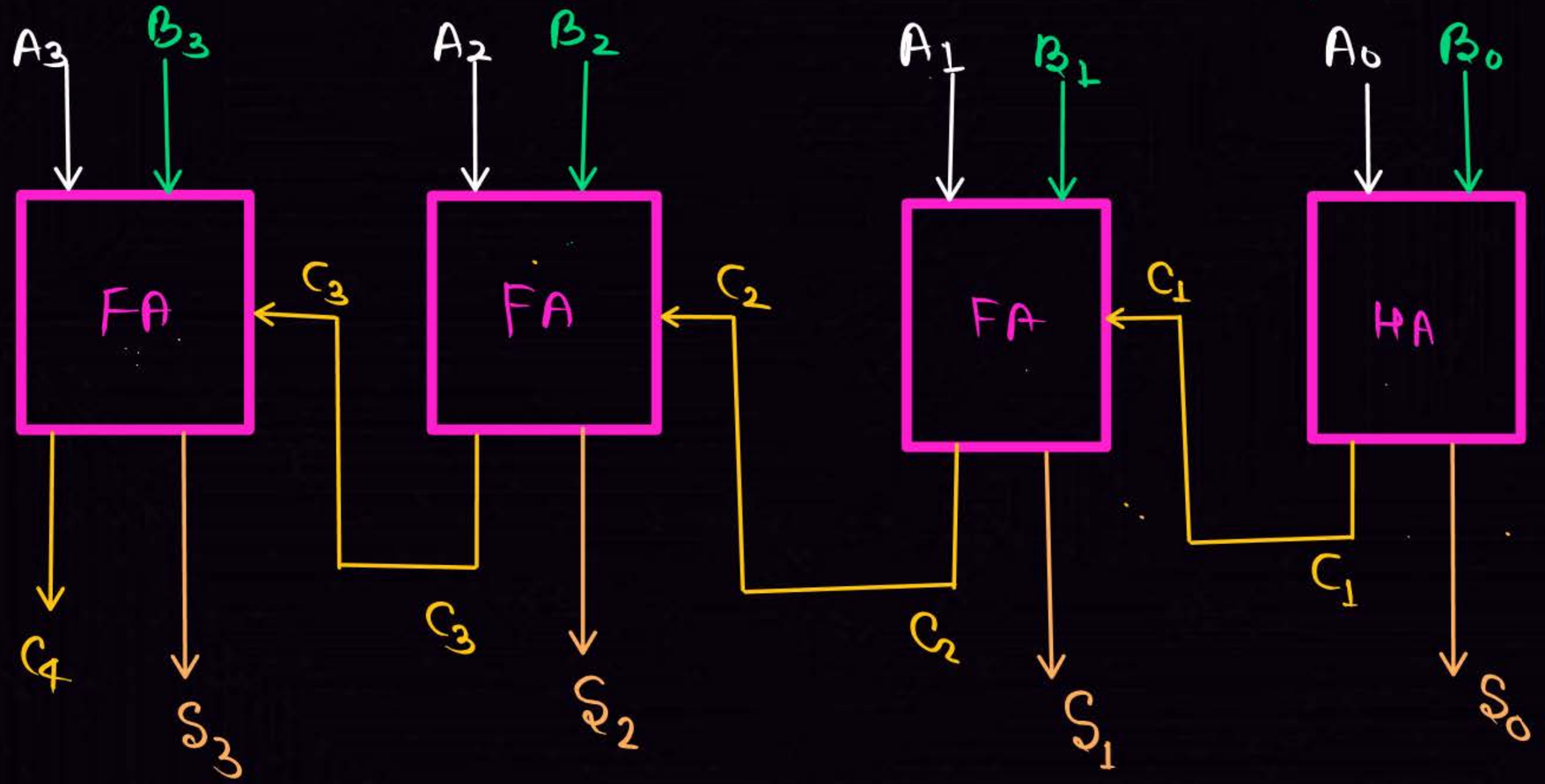
'N' bit parallel Adder

① $(N-1)FA+1.HA$

② 'N' FA

③ $(2N-1)HA+(N-1)$ OR GATE

RIPPLE CARRY ADDER



Worst case Delay

$$T = (N-1)T_{carry} + Max\{T_{sum}, T_{carry}\}$$

Ex $N=4$ $T_{sum}=10s$ $T_{carry}=5s$

$$T = (4-1)\times 5 + Max\{10, 5\}$$

$$= 3\times 5 + 10$$

$$= 25 \text{ Second}$$

$\frac{\text{addition}}{\text{second}}$ or Rate of addition or frequency

$\frac{1}{T}$

Q A 16 bit Ripple carry adder required 12 μs to produce carry and 15 μs to produce sum. Total time required for complete operation, also calculate number of addition per second.

Sol: $T = (n-1)T_{carry} + Max\{T_{sum}, T_{carry}\}$

$n = 16$

$T_{carry} = 12\ \mu s$

$T_{sum} = 15\ \mu s$

$T = (16-1) \times 12\ \mu s + 15\ \mu s$

$T = 195\ \mu s$

$195\ \mu s \longrightarrow$ addition

$1\ s \longrightarrow \frac{1}{195\mu}$

$\frac{1}{T} = \frac{10^6}{195} = \frac{1000 \times 10^3}{195}$ add/sec

$= 5128$ add/sec

Question (NAT)

In a 4-bit parallel adder, sum delay and carry delay of each full adder is t_s = 1 nsec and t_c = 5 nsec respectively. No. of additions performed per sec by adder is N_1. If delays are interchanged then no. of additions performed per sec is N_2, then N_2/N_1 ____2.5____.

case(1) $n=4$
$t_s=1ns$
$t_c=5ns$

$T=3\times5ns+5ns$

$T=20ns$

$N_1=\frac{1}{T}=\frac{1}{20ns}$

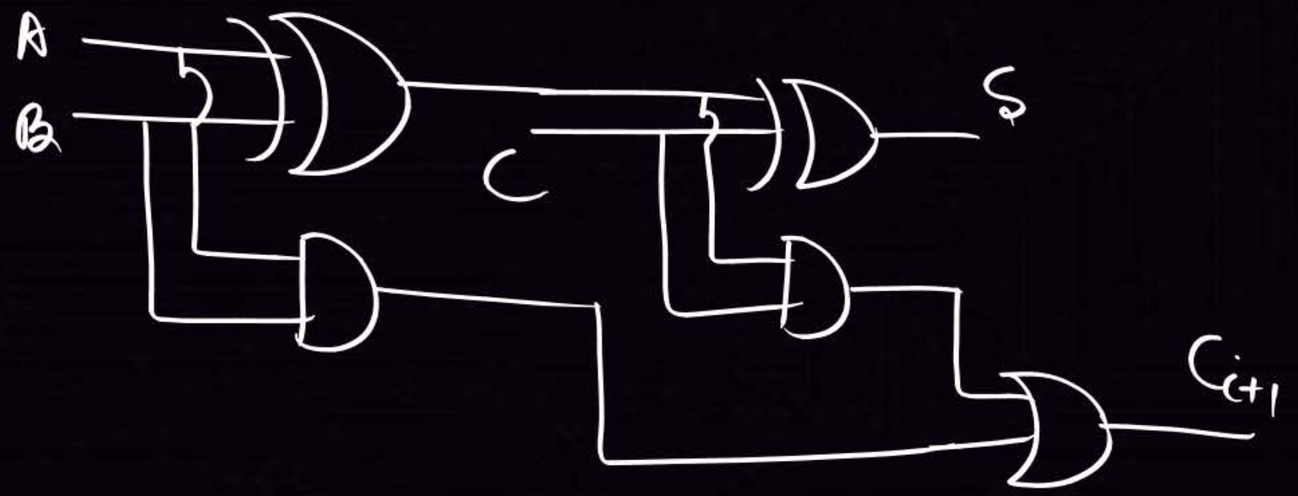
case(2) $n=4$
$t_s=5ns$
$t_c=1ns$

$T=3\times1ns+5=8ns$

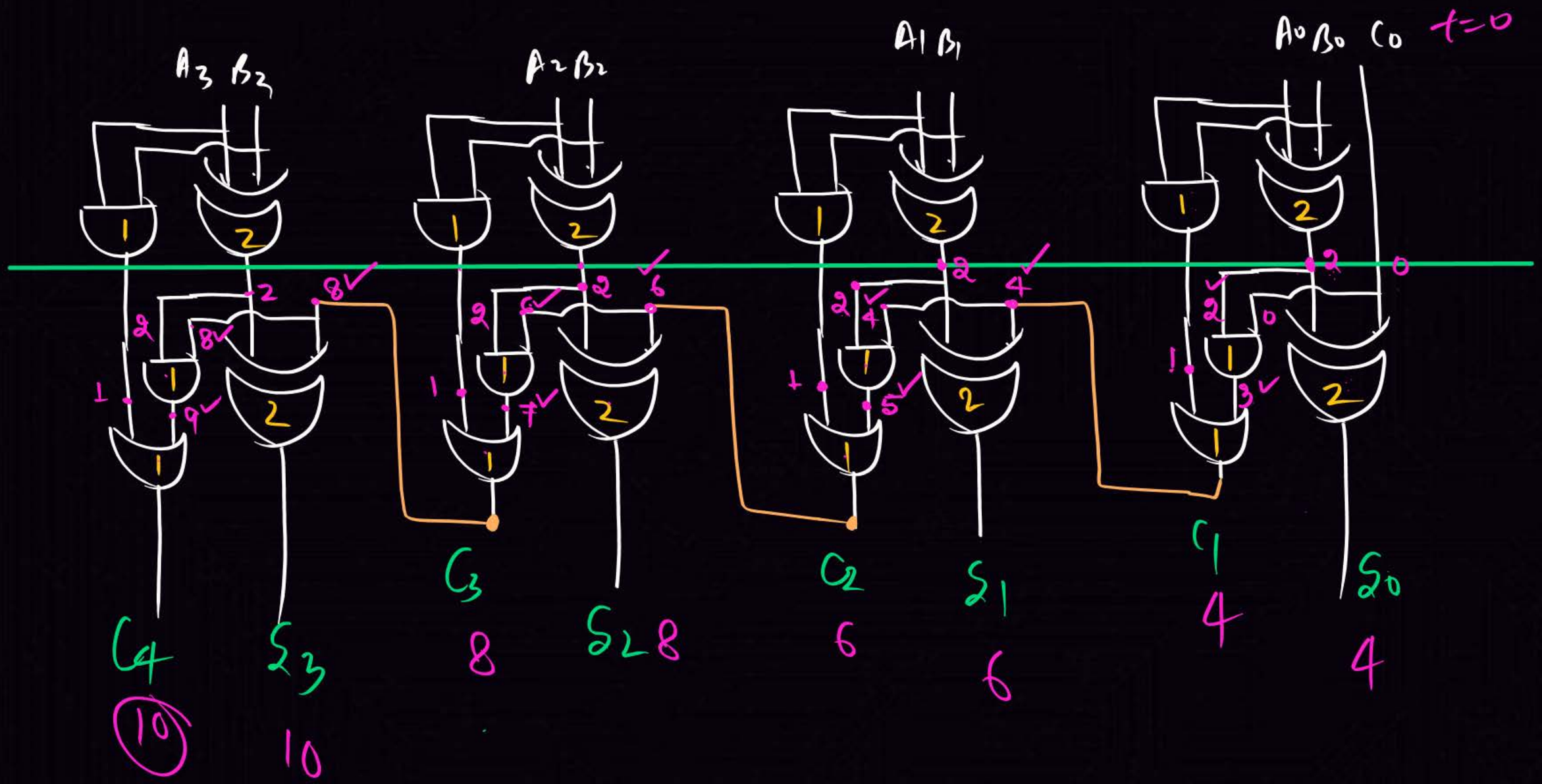
$N_2=\frac{1}{T}=\frac{1}{8ns}$

$\frac{N_2}{N_1}=\frac{\frac{1}{8ns}}{\frac{1}{20ns}}=\frac{20}{8}=2.5$

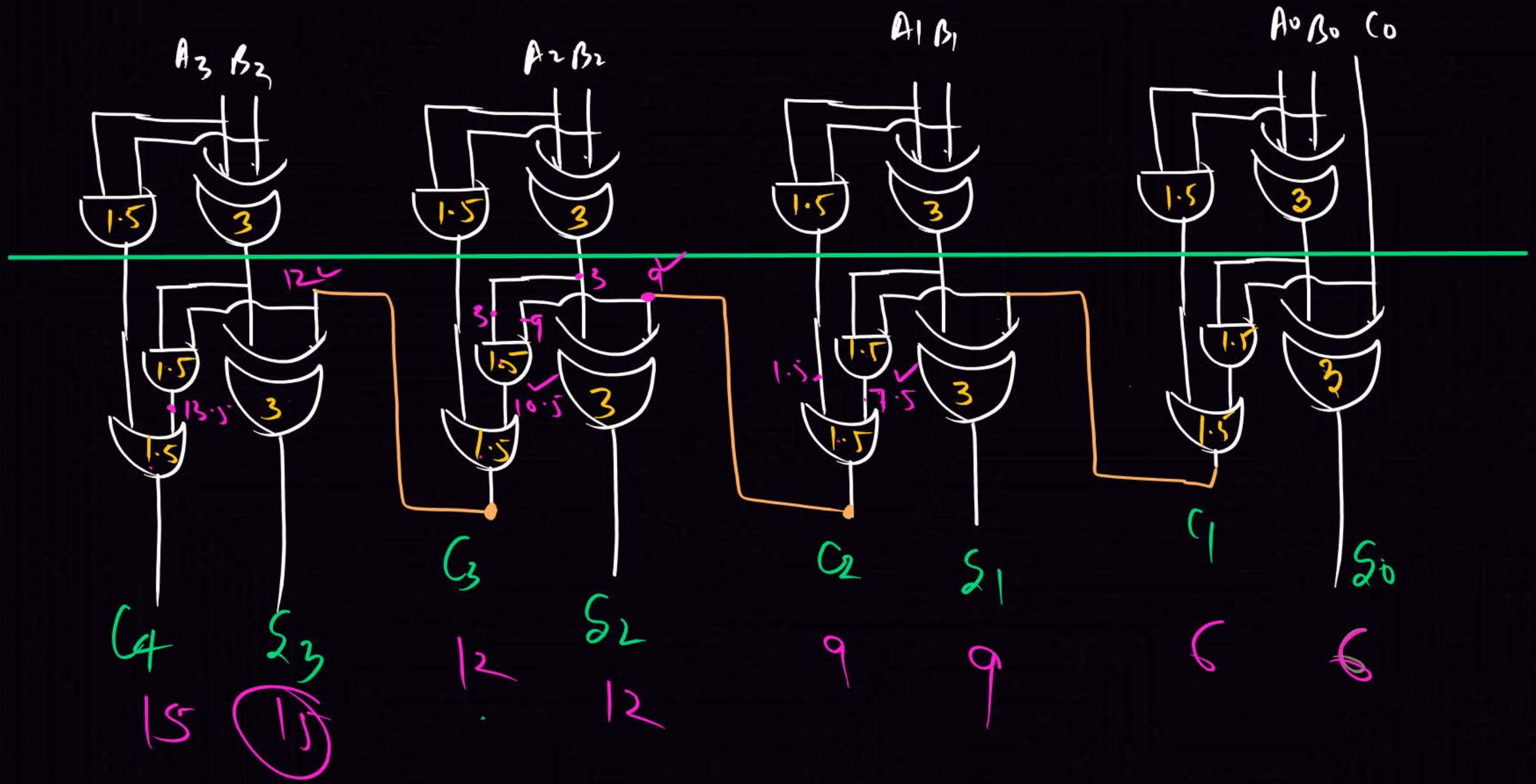
Full adder



Internal circuit Diagram



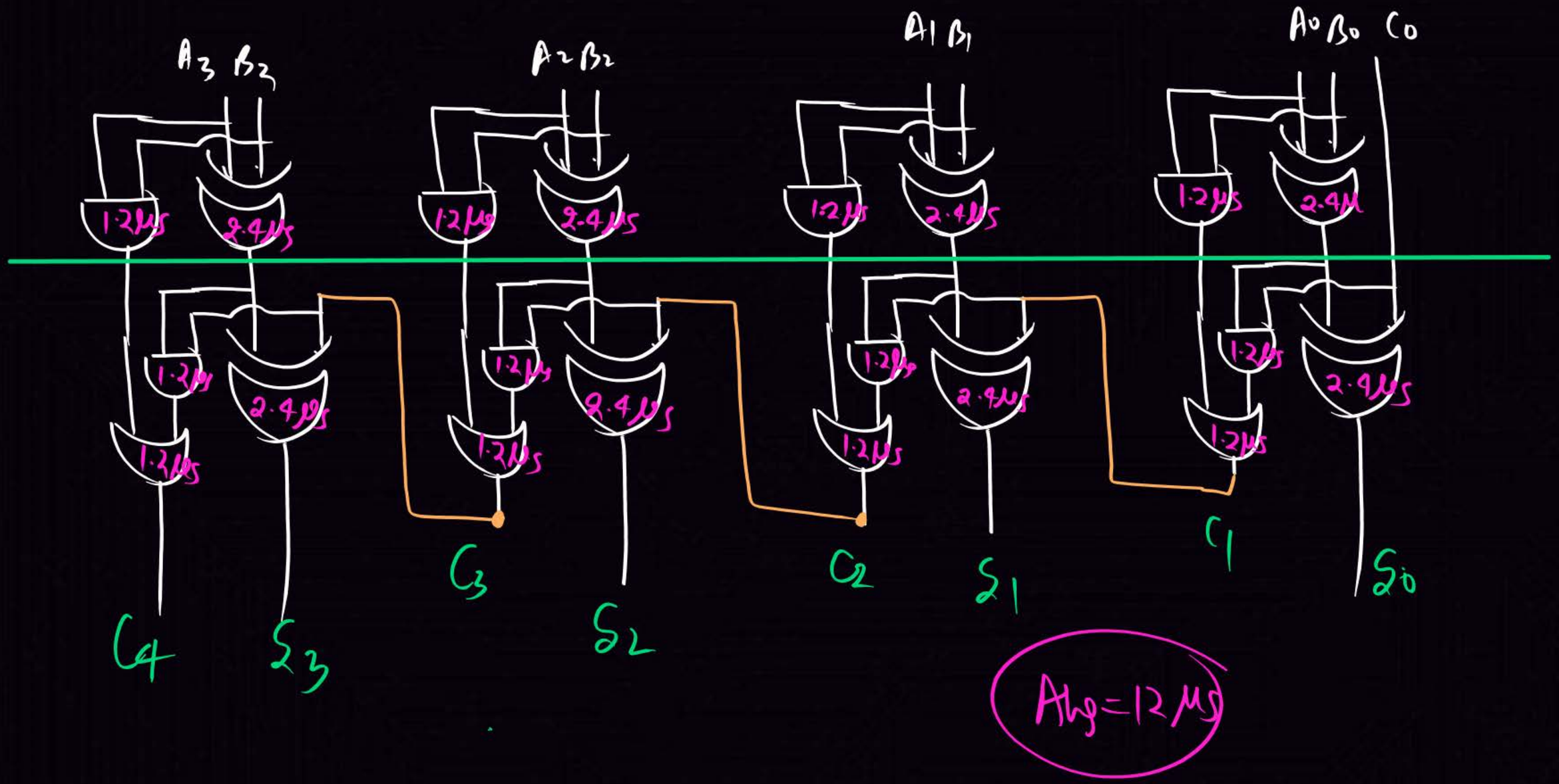
Internal circuit Diagram



NOTE :→ Full adder are designed by HA and OR, & Delay of X-OR GATE, AND GATE, OR GATE is given-

$$T = (n-1)\{\underbrace{T_{AND} + T_{OR}}_{T_{carry}}\} + \underbrace{2 \times T_{XOR}}_{T_{sum}}$$

Internal circuit Diagram



$A_{avg} = 12\,\mu s$

Question **(NAT)**

We have a 4-bit adder in which each full adder has sum delay of t_s = 4 nsec and carry delay of 2 nsec.

In one minute number of additions completed by this full adder is _____ $\times 10^9$.

6

Ans

$n = 4$

$t_s = 4ns$

$t_c = 2ns$

$T = (n-1)T_c + Max\{T_s, T_c\}$

$T = 3 \times 2ns + 4ns$

$T = 10ns$

1 second ⟶ 0.1×10^9 addition

60 second ⟶ $60 \times 0.1 \times 10^9$ addition

$= 6 \times 10^9$ additio

Question **(MCQ)**

Eight 1-bit full adders are cascaded. Each 1-bit full adder generates carry out bit in 10 ns and sum bit in 30 ns. The number of addition performed per second, will be

A 10^9

B 0.5×10^7

C 10^7

D 0.125×10^7

Question **(MCQ)**

#Q. A 16-bit parallel adder is designed using 16, 1-bit full adders. Each 1-bit full adder is designed using X-OR, AND and OR gates. The delay contributed by X-OR gate is 20 ns while that contributed by AND/OR gate is 10 ns. The number of additions per second that a 16-bit parallel adder can perform reliably, will be

A 2.94×10^6

B 3.125×10^6

C 3.33×10^6

D 4×10^6

Question **(MCQ)**

#Q. In a 4-bit carry look ahead adder, the X-OR gate has delay of 10ns and AND/OR gate has delay of 5ns each. The sum will appear after delay of

A 35ns

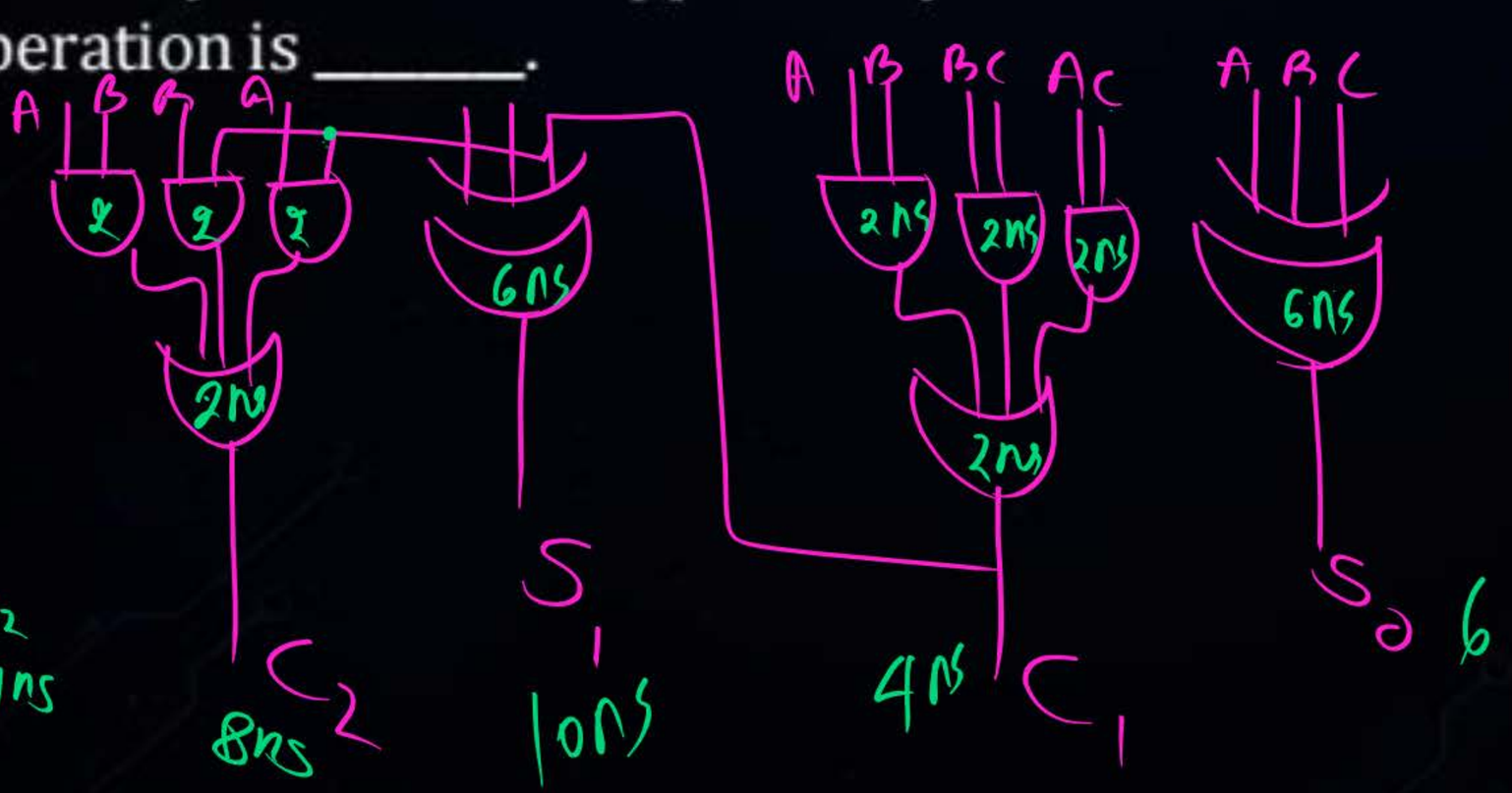
B 30ns

C 25ns

D 20ns

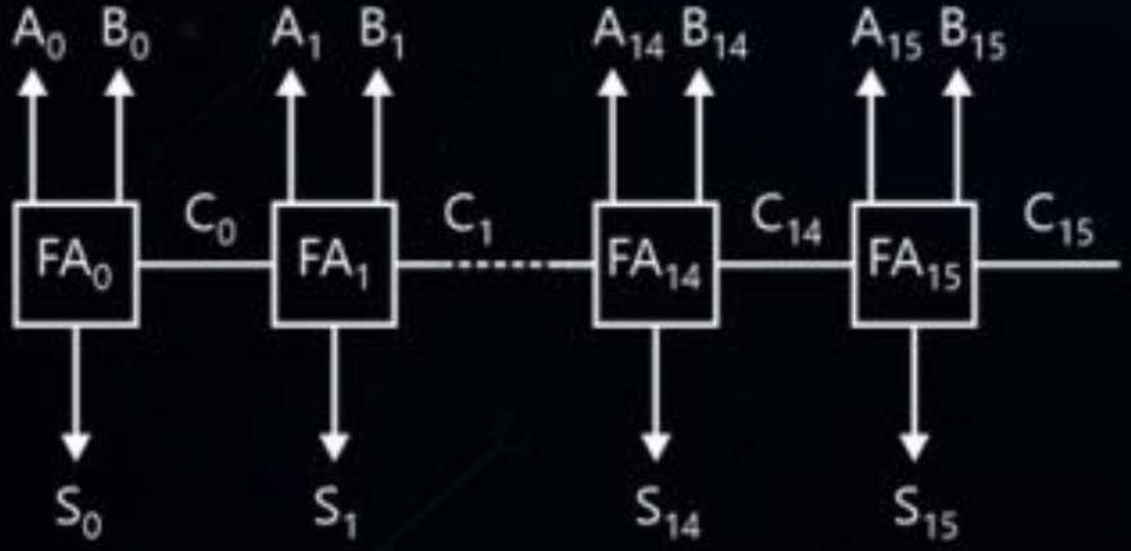
Question **(MCQ)**

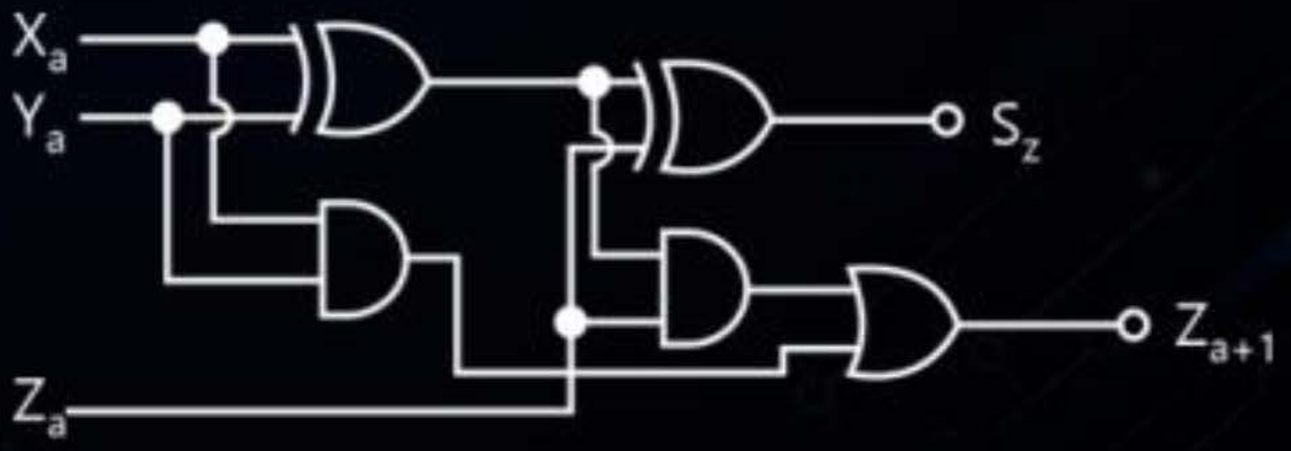
#Q. Each full adder in a 4-bit ripple carry adder, is implemented using one 3-input XOR gate, three 2-input AND gates, and one 3-input OR gate. Let the delay of each 3-input XOR gate be 6ns, that of each 2-input AND gate be 2ns, and that of each 3-input OR gate be 2ns. The total time (in ns) taken by the 4-bit ripple carry adder to do a successful addition operation is _____.



Question **(MCQ)**

A 16-bit ripple carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry-propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be _____ .





Question **(MCQ)**

A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit ripple-carry binary adder is implemented by using full adders. The total propagation time of this 4-bit binary adder in microseconds is,

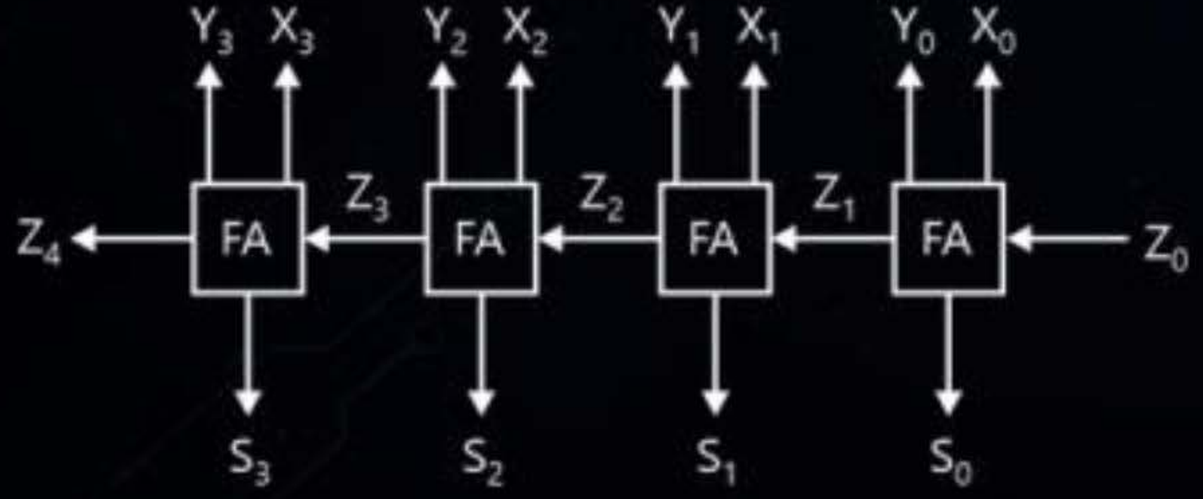


Figure-I

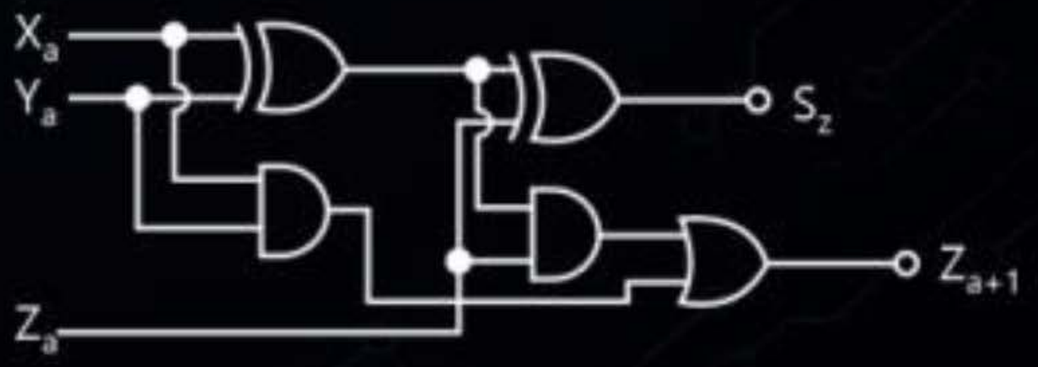


Figure-II

Look Ahead Carry Adder

→ It is the fastest adder among all the adder.

Thank you DOSTO